MSA Pairformer

Expanding protein language modeling to protein–protein interactions

Akiyama, Zhang, Tang, Kim, Mirdita, Steinegger & Ovchinnikov — *Cell* 189, 4964–4979.e8 (2026)

Journal Club

Background

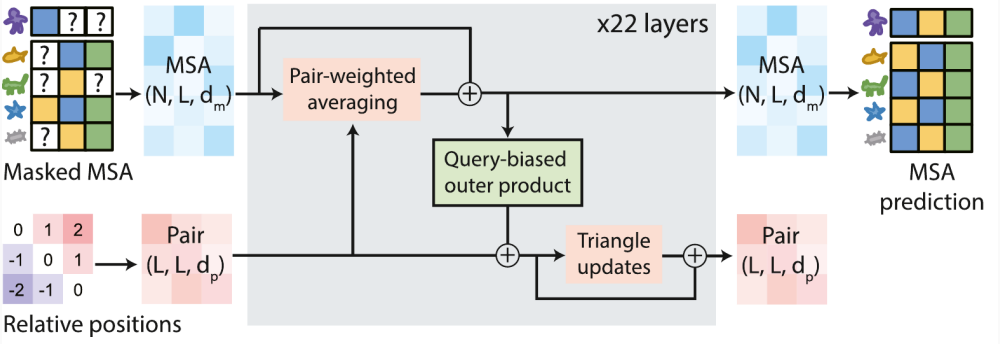
1. **Single-sequence pLMs** (ESM2, ESMC, ...)
   - Read many individual sequences, learn their dependencies, and store the patterns in the parameters; follow scaling laws (more compute, parameters and data $\Rightarrow$ better performance)

2. **MSA-based models** (MSA Transformer, ...)
   - Augment each query with its MSA and build the representation from the coevolution in that alignment

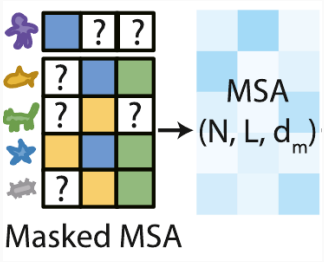
3. **MSA Pairformer** (this paper): a follow-up in MSA-based modeling
   - Uses the AlphaFold3 MSA module, well suited to modeling residue–residue interactions
   - Adds a query-biased outer product: the model adapts to each query, so subfamilies of one protein family with distinct interaction interfaces are handled separately, giving better performance in their experiments

111M parameters **1 H100 $\times$ 10.5 days** trained on monomer MSAs

Architecture



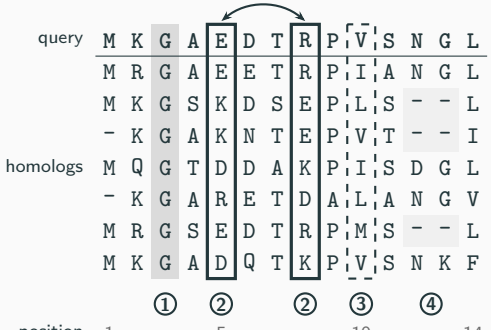
MSA representation



$\mathbf{m} \in \mathbb{R}^{N \times L \times d_m}$: one vector per (sequence, position)

- N = # sequences: row 1 = **query**, other rows = homologs (MMseqs2 + `hhfilter`); 512 in contact benchmarks; up to 4,096 in other analyses, $\leq$320 in training
- L = # aligned positions = **query length** (insertions removed); varies per protein, cropped to 312/320 in training
- d_m = **464** channels, the same in all 22 layers
- **Initialization**: one-hot over 28 tokens $\rightarrow \mathrm{Linear}(28 \rightarrow 464)$, i.e. an embedding lookup
  20 aa + X, B, Z, U, O + gap + `<pad>` + `<mask>`

A toy example of MSA

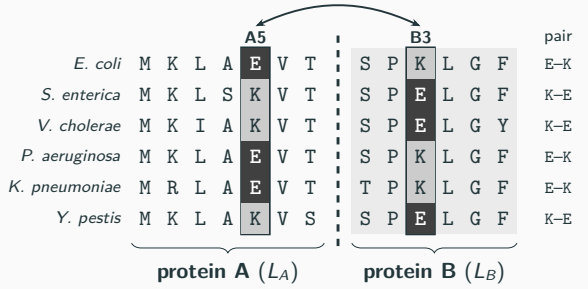


one sequence = one row; the MSA adds the columns

1. **Conservation**: column 3 is G in every homolog ⇒ functionally or structurally essential
2. **Coevolution**: columns 5 and 8 change *together* (E–R, K–E, D–K, R–D), so the charges stay complementary ⇒ likely in contact (salt bridge)
3. **Allowed substitutions**: column 10 varies but stays hydrophobic (V/I/L/M) ⇒ likely buried in the core
4. **Gaps**: positions 12–13 are missing in some homologs ⇒ a variable surface loop

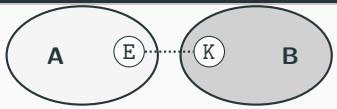
Per-column signals (1, 3, 4) live in $\mathbf{m}$; the pairwise signal (2) is what $\mathbf{z}_{ij}$ is built to capture.

MSA for a pair of proteins



each row = protein A ‖ its partner B *from the same species*

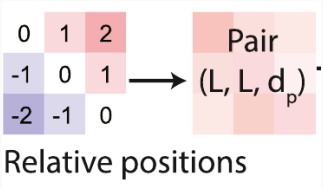
■ E (negative) ■ K (positive): only E–K or K–E, never E–E or K–K



interaction interface

- A mutation at A5 is compensated at B3 (E–K ↔ K–E)
- Same signal as within one protein, now **across** the A | B junction ⇒ an **inter-protein contact**
- **Paralogs**: one genome can carry several copies of A and B, so "same species" is ambiguous; wrong pairs wash out the signal ⇒ pair by **genomic proximity** (operons) or **optimize the pairing** (MP-EM)

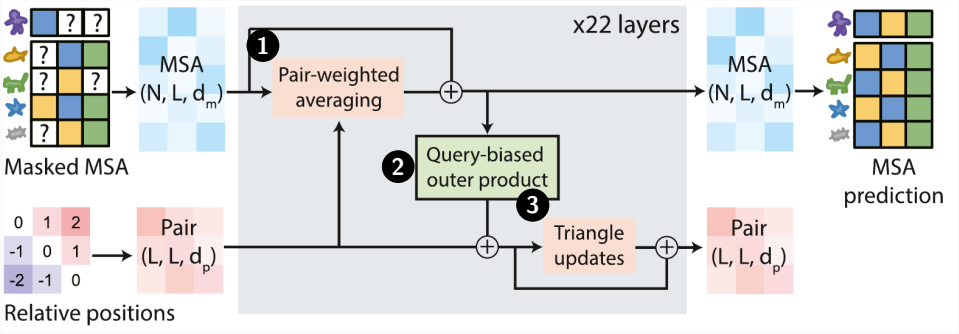
Pair representation



$\mathbf{z} \in \mathbb{R}^{L \times L \times d_p}$: one vector per residue pair (i, j)

- $d_p = 256$ channels, unchanged through all 22 layers
- **Initialization**: one-hot of $\mathrm{clip}(j - i, \pm 32) \rightarrow$ Linear, plus a bonded-neighbor feature ($|i - j| = 1$)
- **Meaning**: $\mathbf{z}_{ij}$ = how residue i relates to residue j (coevolution, contact). It is directional, so $\mathbf{z}_{ij} \neq \mathbf{z}_{ji}$: e.g. $\mathbf{z}_{ij}$ sets how much i reads from j in pair-weighted averaging, and $\mathbf{z}_{ji}$ the reverse. Diagonal $\mathbf{z}_{ii} \approx$ profile of column i
- **Complex A+B**: $L = L_1 + L_2$; inter-chain pairs get the maximum relative distance
- **Read-out**: layer-15 $\mathbf{z} \rightarrow C_\beta$ contact head / Potts head (ConFind head: layer 18)

Architecture: one layer in three steps ($\times 22$)



1 $\mathbf{m} \leftarrow \mathbf{m} + \mathrm{PWA}(\mathbf{m}, \mathbf{z})$

pair-weighted averaging

2 $\mathbf{z} \leftarrow \mathbf{z} + \mathrm{OPM}(\mathbf{m})$

query-biased outer product

3 $\mathbf{z} \leftarrow \mathbf{z} + \mathrm{Tri}(\mathbf{z})$

triangle updates

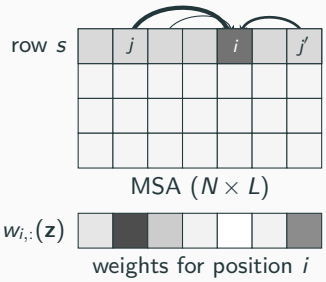
Op 1 Pair-weighted averaging ($\mathbf{z} \to \mathbf{m}$)

$$\boxed{\mathbf{m} \leftarrow \mathbf{m} + \mathrm{PWA}(\mathbf{m}, \mathbf{z})}$$

Simplified form (one head, gating omitted):

$$\mathbf{m}_{si} \leftarrow \mathbf{m}_{si} + \sum_j \underbrace{w_{ij}(\mathbf{z})}_{\text{from pair}} \underbrace{v(\mathbf{m}_{sj})}_{\text{from MSA}}$$

- $v(\cdot)$: transform each MSA entry on its own
- $w_{ij}(\mathbf{z})$: how much position i reads from position j; weights over j sum to 1
- Each sequence reads only from **its own row**, with the **same** weights for all N rows



Intuition: like attention, but *who to listen to* comes from $\mathbf{z}$ (what the model knows about related positions), not from comparing residues.

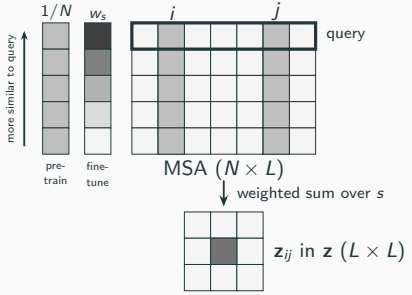
Op 2 Query-biased outer product ($\mathbf{m} \to \mathbf{z}$)

$$\mathbf{z} \leftarrow \mathbf{z} + \mathrm{OPM}(\mathbf{m})$$

Simplified form:

$$\mathbf{z}_{ij} \leftarrow \mathbf{z}_{ij} + f\Big(\sum_s w_s\, a(\mathbf{m}_{si}) \otimes b(\mathbf{m}_{sj})\Big)$$

- $a(\cdot), b(\cdot)$: transform each MSA entry on its own into a small vector
- $a(\mathbf{m}_{si}) \otimes b(\mathbf{m}_{sj})$: what sequence s has at i *together with* what it has at j
- w_s (sums to 1): $1/N$ in pre-training; in fine-tuning, learned per sequence relative to the query, to determine its relevance to the query
- $f(\cdot)$: maps the weighted sum into $\mathbf{z}$



Intuition: with one-hot a, b and $w_s = 1/N$, the sum is the pairwise amino-acid frequency table of classical coevolution analysis. OPM is a learned, query-weighted version of it.

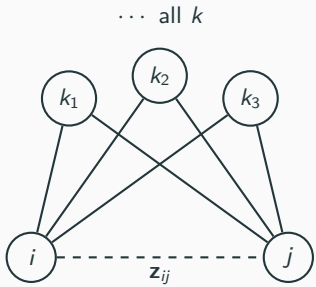
Op 3 Triangle updates ($\mathbf{z} \to \mathbf{z}$)

$$\boxed{\mathbf{z} \leftarrow \mathbf{z} + \mathrm{Tri}(\mathbf{z})}$$

Simplified form (outgoing edges, gating omitted):

$$\mathbf{z}_{ij} \;\leftarrow\; \mathbf{z}_{ij} \;+\; f\Big(\sum_{k} a(\mathbf{z}_{ik}) \odot b(\mathbf{z}_{jk})\Big)$$

- $a(\cdot), b(\cdot)$: transform each pair entry on its own; $f(\cdot)$ maps the sum back into $\mathbf{z}$
- $\sum_k$: for pair (i, j), combine its edges to **every third residue** k
- A second update does the same with incoming edges $\mathbf{z}_{ki}, \mathbf{z}_{kj}$
- Only $\mathbf{z}$ is read; the MSA is not involved



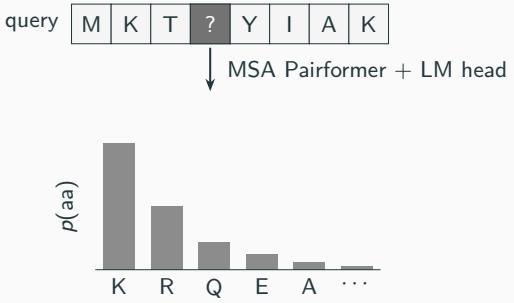
Intuition: check i–j against every third residue k. Strong i–k and j–k can strengthen or weaken i–j, making all pairs mutually consistent and helping separate direct from indirect (transitive) correlations.

Training: two stages, masked-token loss only

	Stage 1: pre-training	**Stage 2: query-biased fine-tuning**
Masking	15% of tokens in *all* rows	15% of tokens in the *query row only*
Seq. weights	uniform $w_s = 1/N$	learned w_s
Query	–	random row ($\geq$90% coverage) each epoch
MSA size	$N \leq 256$ (≥ 8), crop $L = 312$	$128 \leq N \leq 320$, crop $L = 320$
Steps / batch	50k / 12	18k / 32

- Data: **269,455** UniClust30 MSAs (OpenProteinSet), monomers only, no structures
- **Stage 2**: masking only the query rewards finding the rows that are *most informative for that particular sequence*
- The backbone uses sequence-only supervision; contact probes are fitted afterwards on labeled structures.

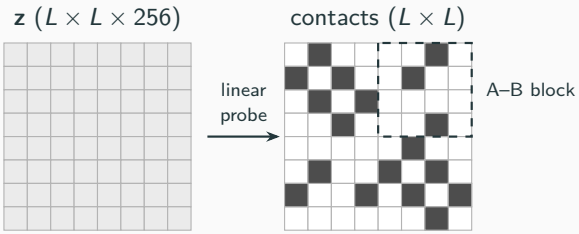
Head 1: LM head (m → amino-acid probabilities)



Intuition: "which amino acid fits here, given the rest of the sequence and its family?"

- Trained together with the model (the masked-token objective)
- **Variant effect**: mask position i, then score $= \log p(\text{mut}) - \log p(\text{wt})$
- **Binder scoring**: sum this over the mutated positions

Head 2: Contact head ($\mathbf{z} \rightarrow$ contact map)



Intuition: if a simple **linear** model can read contacts out of **z**, the structure is already encoded there, without any structure supervision.

- Fit afterwards on only 20 proteins, model frozen
- For a complex A+B: the A–B block gives the **interface** contacts

Inference walkthrough: a new protein comes in

1. **Find homologs.** MMseqs2 search (e.g. ColabFold server / UniRef) $\rightarrow$ A3M alignment. Drop insertion columns so every row aligns to the query's L columns.
2. **Filter / subsample** with `hhfilter`: min. identity to query (15–30%), min. coverage (50–70%), max. pairwise identity (90–95%); cap at $N \approx 512$.
3. **Tokenize**: query is row 1 $\rightarrow$ one-hot tensor $(1, N, L, 28)$.
4. **One forward pass** (bf16, no gradients).
5. **Read out what you need**:
   - contacts: $\mathbf{z}^{(15)} \rightarrow$ contact head $\rightarrow L \times L$ probabilities
   - embeddings: $\mathbf{m}$ (query row) or $\mathbf{z}$ at any layer
   - sequence weights w_s per layer (subfamily analysis)

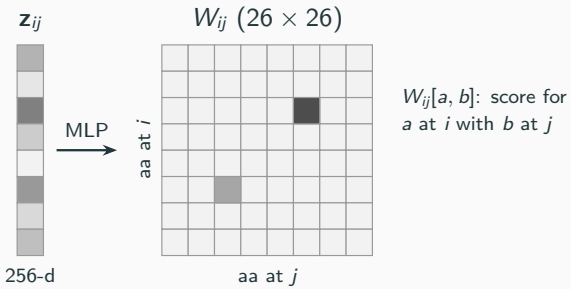
Complex A+B: build a *paired* MSA (rows matched by species / genomic proximity), concatenate $\rightarrow L_1 + L_2$, pass the chain-break index.

How it differs from earlier MSA models

	MSA Transformer	**AF3 MSA module**	**MSA Pairformer**
Representations	MSA only	MSA + pair	MSA + pair
Mixing across sequences	tied row + column attention	uniform outer-product mean	query-biased weighted outer product
Triplet reasoning	implicit	triangle updates	triangle multiplicative updates
Supervision	masked tokens	structure (inside AF3)	masked tokens only

Core argument: averaging over the whole family can wash out **subfamily-specific** signals such as different interfaces or oligomeric states. Query biasing lets the model choose which rows to listen to.

Head 3: Potts head (z → an amino-acid table per pair)



$W_{ij}[a, b]$: score for a at i with b at j

Intuition: translate **z** into an explicit table of "which amino-acid pairs go together", the same form as classical coevolution models.

- Score any sequence by **looking up** its amino acids in the tables: no new forward pass
- Used to **pair interacting paralogs**: try pairings, keep the highest-scoring one

Using the Potts head: pairing paralogs (MP-EM)



- **E step**: mask 30% of MSA tokens; run MSA Pairformer + Potts head (weights fixed) → new V, W; score each candidate A–B pair within a species (A–B couplings only)
- **M step**: per species, pick the one-to-one pairing with the highest total pseudolikelihood (SciPy matching operator; decaying Gaussian exploration noise)
- Final consensus: average the 400 lowest-loss permutation matrices from 500 steps, then match again

Pairing paralogs: accuracy and computational cost

Five independently sampled alignments per complex; one query pair fixed. HK–RR: 384 sequences; MALG–MALK: 256 sequences.

Method	HK–RR precision	MALG–MALK precision	Mean runtime
MSA Transformer DiffPALM	0.63	0.31	$>$18 min
MP-PDP	0.68	**0.70**	$<$4 min
MP-EM	**0.70**	0.67	$\sim$1 min

- Peak reserved GPU memory: 87/74 GB for DiffPALM versus $<$**3 GB** for both Potts methods (96-GB H100 experiments).
- MALG–MALK interface $P@K$: **0.02** with DiffPALM pairings, **0.44** with MP-EM, **0.47** with MP-PDP.

Cell 2026, Fig. 3E–G; STAR Methods, Pairing experiments. Precision = fraction of correct sequence pairs.

Interface contacts: transfer from monomers to complexes

Question: does a model trained on individual chains recover inter-chain coevolution?

- 30 conserved bacterial complexes; proximity-filtered paired MSAs; fusion genes removed.
- Comparison with MSA Transformer uses **27 complexes**: three exceed its length limit.
- *P@K*: precision of the top *K* inter-chain predictions; *K* is the number of true interface contacts (C_β distance ≤ 8 Å).

Model	**Median** *P@K*
MSA Pairformer	**0.52**
MSA Transformer	0.19
Inverse covariance	0.09
gLM2	0.03
E1 / ESMC	0.01 / 0.01

$\sim$**2.7-fold** higher precision than MSA Transformer; adjusted $p \leq 0.05$ against all methods.

Cell 2026, Fig. 2A–F, Results pp. 4966–4968; STAR Methods. Backbone frozen; probe uses 20 monomers.

Binder ranking: a partner-context control

ParD3 antitoxin library: **9,194 variants** at four interface positions; **465** have experimental fitness ≥ 0.15 (binders).

Ranking by pseudolikelihood	Precision among top 465
MSA Pairformer, ParD3 + ParE3	**0.74**
MSA Pairformer, ParD3 alone	0.50
MSA Transformer / GREMLIN	0.68 / 0.68
E1 / PoET / ESMC	0.57 / 0.48 / 0.45

Removing the toxin partner reduces precision by **24 percentage points**. AUPRC is 0.82 for Pairformer versus 0.76 for MSA Transformer; AF3 ipSAE ranking precision is 0.53.

Cell 2026, Fig. 3A–D, Results pp. 4968–4969; STAR Methods. Retrospective cellular-growth binding proxy.

Query bias: same alignment, different interface signals

Controlled input: one 4,096-sequence response-regulator alignment; change only which sequence is designated as the query.

Query subfamily	MSA fraction	Uniform weights	Query-biased
OmpR	∼12%	1/19	**8/19**
GerE	∼61%	3/23	**5/23**

Recovered subfamily-specific homodimer-interface contacts.

- Strongest recovery gain occurs for minority OmpR, whose signal uniform averaging dilutes.
- All 516 OmpR sequences and 81% of GerE sequences are up-weighted, including GerE sequences with only 9% identity to the query.

Cell 2026, Fig. 4B–E, Results pp. 4970–4972; STAR Methods, Response regulator analysis.

Single-protein benchmarks: contacts and variant effects

CASP15 long-range contacts

45 targets shorter than 1,024 residues; up to 512 homologs; separation ≥ 24.

Model	**Median** *P@L*
MSA Pairformer	**0.66**
MSA Transformer	0.50
ESM2-15B	0.50

ProteinGym substitutions

217 DMS datasets; average Spearman correlation with experimental measurements:

$$\rho = \mathbf{0.45}$$

Outperforms ESM models and the MSA Transformer 5-MSA ensemble. Reported relative gains: 4% over the ensemble and 8% over ESM2-650M.

Interpretation: strong contact prediction need not sacrifice variant-effect ranking. ProteinGym averaging first groups by UniProt ID within readout type, then averages across readout types.

Cell 2026, Fig. 5A/C, Results pp. 4972–4973; STAR Methods, ProteinGym variant effect prediction.

New sequence data can improve predictions without retraining

Intervention: append homologs found in the Logan sequence database to existing UniRef alignments; retain the trained model weights.

CASP15 median long-range *P@L*	**UniRef**	**+ Logan**
MSA Pairformer	0.66	**0.71**
MSA Transformer	0.50	**0.55**

- Viral variant-effect prediction improves from mean Spearman $\rho = 0.471$ to 0.487 for Pairformer.
- Both MSA models also recover contacts from synthetic alignments of novel designed folds; this is evidence for **inference-time extraction** beyond training examples.

Cell 2026, Fig. 5A–C, Results pp. 4972–4973; STAR Methods, Logan augmentation / out-of-distribution proteins.

Limits and journal-club discussion

1. **The MSA is part of the predictor.** Sparse homologs, disordered regions, distant alignments, and rearrangements can break its assumptions. Query bias can reduce effective depth when all sequences share relevant signal.

2. **Pairing is not yet general.** Two bacterial benchmark interactions; equal-size MSAs and one-to-one matching assumed. Highly conserved interfaces, unequal paralog counts, eukaryotic and cross-species interactions remain open problems. Genomic proximity does not transfer directly to eukaryotes.

3. **Performance is not prospective design validation.** A retrospective toxin-antitoxin library does not establish affinity calibration or proteome-wide discovery. Parameter count also omits alignment search and deep-MSA inference cost. Which held-out families and negative controls would test those claims?

Cell 2026, Discussion / Limitations, pp. 4974–4975. Final questions and scope interpretation are ours.

Sources and verification scope

1. Akiyama et al., Cell 189(16), 4964–4979.e8 (2026): full 25-page journal PDF supplied by the user, including STAR Methods, read for this revision. Newly added benchmark slides use the final published results and figure numbers.

2. Author code / Cell 2026 analysis notebooks and analysis data: reproducibility resources. MP-EM is checked against STAR Methods, Algorithm 2.

3. Separate journal Methods S1 / supplementary tables were not supplied or retrieved. The earlier 2025 preprint v1 and its Appendix were read, but their numerical benchmarks have been superseded here.

The original pedagogical architecture and MP-EM walkthrough is retained; published evidence and limitations complete the discussion.